
Software Requirements Specification

for

**VisionEdge: Latency-Optimized Local Perception &
TTS Pipeline**

Version 1.0

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Revision History

Author Name	Date	Description	Version
Madhav Juneja	14/1/26	Initial draft of Software Requirements Specification covering system overview, functional and non-functional requirements	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the software requirements for **VisionEdge**, an offline, mobile-based assistive application designed to provide real-time visual perception and audio feedback for visually impaired users. This document defines the functional and non-functional requirements of the system and specifies the scope, behavior, and constraints of the software.

The purpose of this SRS is to serve as a formal reference for stakeholders, including academic evaluators, developers, and project team members, by providing a clear and unambiguous description of the system requirements. This document will be used as a basis for system design, implementation, testing, and evaluation during the Software Engineering course.

1.2 Document Conventions

This document follows standard **IEEE Software Requirements Specification (SRS)** conventions. Section and subsection numbering adhere to the structure defined in the SRS template. Requirements are stated using clear and precise language, with functional requirements expressed using the term “**shall**” to indicate mandatory system behavior.

Bullet points and tables are used where appropriate to improve clarity and readability. Acronyms and abbreviations are expanded upon first use and defined in a separate glossary. All requirements are written to be concise, verifiable, and necessary.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- Course instructors and teaching assistants responsible for evaluating the project
- Developers and project team members involved in system design and implementation
- Testers and reviewers responsible for validation and verification activities

Readers are advised to begin with **Section 1** to understand the purpose and scope of the system. **Section 2** provides a high-level overview of the product and its operating environment. **Section 3** describes the external interfaces of the system. Subsequent sections define detailed system features and non-functional requirements.

1.4 Product Scope

VisionEdge is a software system that converts a smartphone into a real-time assistive visual aid. The system captures live video from the device camera, performs on-device visual perception to detect objects and scene elements, and generates immediate spoken feedback using a lightweight neural Text-to-Speech (TTS) pipeline.

The primary goals of VisionEdge are to:

- Enable real-time environmental awareness for visually impaired users
- Ensure low-latency response suitable for continuous use
- Preserve user privacy by operating entirely offline
- Provide an energy-efficient and accessible mobile solution

VisionEdge is intended for everyday use in indoor and outdoor environments and does not rely on cloud-based processing or internet connectivity.

1.5 References

The following documents and resources are referenced in the preparation of this SRS:

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*
- Course materials provided as part of the Software Engineering curriculum
- Technical documentation related to mobile application development and on-device machine learning

2. Overall Description

2.1 Product Perspective

VisionEdge is a **new, standalone mobile application** developed as part of a Software Engineering academic project. It is designed as an independent assistive system and does not replace or extend any existing commercial product. The system operates entirely on a smartphone and performs all processing locally on the device.

From a system perspective, VisionEdge consists of multiple interconnected modules, including a camera input module, a vision perception module, a text generation component, a neural text-to-speech (TTS) pipeline, and a user interaction layer. These components work together to transform real-time visual input into spoken audio feedback. The system relies on the underlying mobile operating system for access to hardware resources such as the camera, audio output, and processing units.

VisionEdge functions as a self-contained application and does not require interaction with external cloud services or network-based systems during normal operation.

2.2 Product Functions

At a high level, the VisionEdge system provides the following functions:

- Capture real-time video input from the smartphone camera
- Detect objects and scene elements from captured video frames
- Generate textual descriptions from detected visual information
- Convert textual descriptions into spoken audio using an offline TTS pipeline
- Deliver low-latency audio feedback to the user
- Support accessible user interaction suitable for visually impaired users

Detailed functional requirements are specified in later sections of this document.

2.3 User Classes and Characteristics

The primary user class for the VisionEdge system is:

- **Visually Impaired Users:**

Users with partial or complete visual impairment who require auditory assistance to understand their surrounding environment. These users may have limited ability to interact with complex graphical interfaces and therefore require simple, accessible, and audio-driven interaction mechanisms.

Secondary user classes include:

- **Developers and Testers:**

Individuals involved in developing, testing, and evaluating the system. This user class interacts with the system primarily during development and testing phases and is not the target end-user of the application.

The system is primarily optimized to satisfy the needs of visually impaired users.

2.4 Operating Environment

VisionEdge operates within the following environment:

- Smartphone devices equipped with a camera, speaker, and touch interface
- Mobile operating systems supporting on-device machine learning inference
- On-device processing units such as CPU, GPU, or Neural Processing Units (NPUs)
- Offline operation without dependency on internet connectivity

The system must coexist with other applications on the device without significantly degrading overall device performance or user experience.

2.5 Design and Implementation Constraints

The VisonEdge system is subject to the following constraints:

- The system must operate entirely offline, without reliance on cloud-based services
- Processing must be optimized to meet low-latency requirements on mobile hardware
- Vision and TTS models must be lightweight to reduce memory and battery usage
- Development must follow standard software engineering practices as required by the course
- The system must ensure accessibility and ease of use for visually impaired users

These constraints limit the choice of technologies, models, and design approaches used in system implementation.

2.6 User Documentation

The following user documentation will be provided with the VisonEdge system:

- A brief user guide describing application usage and basic interactions
- In-application audio instructions to assist users during operation
- Developer documentation describing system architecture and module behavior

Documentation will be designed to be concise and accessible.

2.7 Assumptions and Dependencies

The VisonEdge project is based on the following assumptions and dependencies:

- Users possess a smartphone with sufficient computational capability to support on-device inference
- The mobile operating system provides stable APIs for camera access and audio output
- Adequate lighting conditions are available for effective visual perception
- Pre-trained and optimized machine learning models perform reliably across common environments
- System performance may vary depending on device hardware specifications

3. External Interface Requirements

3.1 User Interfaces

The VisionEdge system provides a mobile user interface designed with accessibility as the primary focus. The interface minimizes visual dependency and relies heavily on audio feedback to communicate system status and results to the user.

User interaction is supported through simple touch-based controls and large interface elements. Audio cues are used to confirm user actions, indicate errors, and provide feedback. The interface avoids complex layouts and ensures consistent interaction patterns to improve usability for visually impaired users.

3.2 Hardware Interfaces

VisionEdge interfaces with the following hardware components of the smartphone:

- **Camera:** Used to capture real-time video input for visual perception
- **Speaker or Headphones:** Used to deliver synthesized audio output
- **Touchscreen:** Used for basic user interaction and control

All hardware interactions are managed through standard mobile operating system interfaces.

3.3 Software Interfaces

The system interfaces with internal software components and system-level services, including:

- Mobile operating system APIs for camera access, audio playback, and resource management
- On-device machine learning inference frameworks for executing vision and TTS models
- Application-level modules for vision processing, text generation, and audio synthesis

All software interactions occur locally on the device, and no external online software services are required

3.4 Communications Interfaces

VisionEdge does not require network-based communication interfaces for normal operation. The system does not transmit or receive data over the internet and does not depend on communication protocols such as HTTP or FTP.

Optional communication interfaces may be used only for application installation, updates, or debugging purposes, but these are not part of the core system functionality.

4. System Features

This section describes the major system features provided by VisionEdge. Each feature represents a core service offered by the system and is defined with its description, priority, stimulus/response behavior, and detailed functional requirements.

4.1 Real-Time Visual Perception

4.1.1 Description and Priority

This feature enables the system to capture live visual input from the smartphone camera and analyze it to detect objects and scene elements in the user's environment. It forms the foundation of the assistive functionality provided by VisionEdge.

Priority: High

4.1.2 Stimulus/Response Sequences

- **Stimulus:** The user launches the application and activates visual assistance.
- **Response:**

1. The system initializes the camera.
2. The system continuously captures video frames.
3. The system processes the frames using on-device vision models to detect objects and scene elements.

4.1.3 Functional Requirements

- **REQ-1:** The system shall activate the smartphone camera when visual assistance is enabled.
- **REQ-2:** The system shall capture video frames in real time for analysis.
- **REQ-3:** The system shall perform object and scene detection using on-device vision models.
- **REQ-4:** The system shall handle camera access errors and notify the user through audio feedback.

4.2 Textual Description Generation

4.2.1 Description and Priority

This feature converts detected visual information into concise textual descriptions that can be used for audio narration. The descriptions are designed to be meaningful, context-aware, and suitable for real-time assistance.

Priority: High

4.2.2 Stimulus/Response Sequences

- **Stimulus:** Objects or scene elements are detected in the visual input.
- **Response:**
 1. The system extracts relevant information from detection results.
 2. The system generates a textual description of the environment.
 3. The generated text is forwarded to the TTS module.

4.2.3 Functional Requirements

- **REQ-5:** The system shall convert detected objects and scenes into textual descriptions.
- **REQ-6:** The system shall generate concise and understandable descriptions suitable for audio output.
- **REQ-7:** The system shall update descriptions dynamically as the visual scene changes.

4.3 Offline Text-to-Speech (TTS) Output

4.3.1 Description and Priority

This feature enables the system to convert textual descriptions into spoken audio using an offline neural text-to-speech pipeline. It ensures that audio feedback is generated without internet connectivity.

Priority: High

4.3.2 Stimulus/Response Sequences

- **Stimulus:** A textual description is generated by the system.
- **Response:**
 1. The system processes the text using the TTS module.
 2. The system generates audio output.
 3. The audio is played through the device speaker or connected audio output.

4.3.3 Functional Requirements

- **REQ-8:** The system shall generate speech output from textual descriptions using an offline TTS model.
- **REQ-9:** The system shall deliver audio output through the device speaker or connected audio devices.
- **REQ-10:** The system shall notify the user if audio output fails.

4.4 Low-Latency Processing and Feedback

4.4.1 Description and Priority

This feature ensures that the time taken from visual input capture to audio output is minimized to support real-time assistance. Low latency is critical for usability and safety.

Priority: High

4.4.2 Stimulus/Response Sequences

- **Stimulus:** Continuous visual input is processed by the system.
- **Response:**
 1. The system processes vision and TTS pipelines efficiently.
 2. The system delivers timely audio feedback with minimal delay.

4.4.3 Functional Requirements

- **REQ-11:** The system shall optimize processing pipelines to minimize end-to-end latency.
- **REQ-12:** The system shall prioritize real-time feedback over non-essential processing tasks.

4.5 Accessibility and User Control

4.5.1 Description and Priority

This feature provides accessible interaction mechanisms that allow visually impaired users to easily control system operation, such as starting or stopping visual assistance.

Priority: Medium

4.5.2 Stimulus/Response Sequences

- **Stimulus:** The user performs an interaction to control system operation.
- **Response:**
 1. The system recognizes the user input.
 2. The system performs the requested action.

3. The system confirms the action using audio feedback.

4.5.3 Functional Requirements

- **REQ-13:** The system shall provide simple interaction mechanisms suitable for visually impaired users.
- **REQ-14:** The system shall allow users to start and stop visual assistance.
- **REQ-15:** The system shall provide audio confirmation for user actions.

5. Other Nonfunctional Requirements

This section describes the nonfunctional requirements that define the performance, safety, security, and quality attributes of the VisionEdge system. These requirements specify constraints and characteristics that affect system behavior and user experience but are not directly related to specific system features.

5.1 Performance Requirements

The VisionEdge system is required to meet the following performance requirements to ensure real-time and reliable operation:

- **PERF-1:** The system shall ensure an end-to-end latency of less than **2 seconds** from video capture to audio output under normal operating conditions.
- **PERF-2:** The system shall aim to deliver audio feedback within **1 second** on supported devices with adequate hardware capabilities.
- **PERF-3:** The system shall process visual input continuously without noticeable interruptions during normal use.
- **PERF-4:** The system shall limit excessive CPU, memory, and battery consumption to allow sustained usage on mobile devices.

5.2 Safety Requirements

The VisionEdge system is designed to assist users without causing physical or operational harm. The following safety requirements apply:

- **SAFE-1:** The system shall provide clear and timely audio feedback to reduce the risk of user confusion.

- **SAFE-2:** The system shall notify the user if critical system components such as the camera or audio output become unavailable.
- **SAFE-3:** The system shall avoid generating misleading or unsafe descriptions when visual information is uncertain.

VisionEdge is intended as an assistive tool and does not replace human judgment. Users are expected to remain aware of their surroundings while using the system.

5.3 Security Requirements

The VisionEdge system must preserve user privacy and ensure secure operation:

- **SEC-1:** The system shall operate entirely offline and shall not transmit visual or audio data to external servers.
- **SEC-2:** The system shall not store sensitive visual data unless explicitly required for system operation.
- **SEC-3:** The system shall follow secure coding practices to prevent unauthorized access to system resources.

By performing all processing locally, VisionEdge minimizes exposure to security and privacy risks.

5.4 Software Quality Attributes

The VisionEdge system shall exhibit the following software quality attributes:

5.4.1 Usability

- The system shall provide an accessible interface optimized for visually impaired users.
- Audio feedback shall be clear, concise, and easy to understand.

5.4.2 Reliability

- The system shall continue operating reliably under normal usage conditions.
- The system shall handle errors gracefully and notify the user when failures occur.

5.4.3 Maintainability

- The system shall be designed using modular components to allow easy updates and enhancements.
- Code and system components shall follow standard documentation and coding practices.

5.4.4 Portability

- The system shall be deployable on a range of modern smartphones supporting on-device machine learning.

5.5 Business Rules

The following business rules govern the operation of the VisonEdge system:

- **BR-1:** The system shall provide assistive functionality exclusively for accessibility purposes.
- **BR-2:** The system shall prioritize user privacy and offline operation over feature expansion that requires cloud connectivity.
- **BR-3:** Only authorized users of the device shall be able to operate the application.

6. Other Requirements

This section specifies additional requirements for the VisonEdge system that are not covered elsewhere in this document but are relevant to the successful development and deployment of the software.

6.1 Accessibility Requirements

- The system shall be designed to support accessibility needs of visually impaired users.
- Audio-based guidance and feedback shall be provided for all primary system operations.
- The system shall minimize reliance on visual interaction wherever possible.

6.2 Data Management Requirements

- The system shall process visual data in real time without persistent storage of raw video frames.
- Temporary data generated during processing shall be discarded after use.

6.3 Legal and Ethical Requirements

- The system shall comply with applicable data privacy and ethical guidelines related to assistive technologies.
- User data shall not be shared with third parties.

6.4 Reuse and Extensibility Requirements

- The system architecture shall support future enhancements such as additional perception modules or language support.
- Components shall be designed to allow reuse in related assistive applications.

Appendix A: Glossary

Term	Description
Assistive Technology	Technology designed to support users with disabilities
Edge AI	Execution of AI models directly on a device without cloud dependency
TTS	Text-to-Speech
Latency	Time delay between input capture and output generation
On-device Processing	Computation performed locally on a mobile device
SRS	Software Requirements Specification

Appendix B: Analysis Models

Analysis models such as high-level architecture diagrams, data flow diagrams, and use case diagrams may be developed during the system design phase to further illustrate system behavior and component interactions.

Appendix C: To Be Determined List

The following items are identified as **To Be Determined (TBD)** and will be finalized during later stages of development:

- TBD-1: Exact vision model architecture and parameters
- TBD-2: Final TTS model selection and configuration
- TBD-3: Detailed performance benchmarks across different smartphone devices